

## 37 LinkedList Quicksort for Linked Lists.[Very Important]

```
class Solution {
private:
    Node* getTail(Node* cur) {
        while (cur && cur->next) cur = cur->next;
        return cur;
    }
    // Partition the list around pivot (last node)
    Node* partition(Node* head, Node* end, Node** newHead, Node** newEnd) {
        Node* pivot = end;
        Node *prev = NULL, *cur = head, *tail = pivot;

        while (cur != pivot) {
            if (cur->data < pivot->data) {
                if (!(*newHead)) *newHead = cur;
                prev = cur;
                cur = cur->next;
            } else {
                if (prev) prev->next = cur->next;
                Node* tmp = cur->next;
                cur->next = NULL;
                tail->next = cur;
                tail = cur;
                cur = tmp;
            }
        }
        if (!(*newHead)) *newHead = pivot;
        *newEnd = tail;
        return pivot;
    }
    Node* quickSortRecur(Node* head, Node* end) {
        if (!head || head == end) return head;
        Node *newHead = NULL, *newEnd = NULL;
        Node* pivot = partition(head, end, &newHead, &newEnd);
        // Sort the left part before pivot
        if (newHead != pivot) {
            Node* tmp = newHead;
            while (tmp->next != pivot) tmp = tmp->next;
            tmp->next = NULL;

            newHead = quickSortRecur(newHead, tmp);

            tmp = getTail(newHead);
            tmp->next = pivot;
        }
        // Sort the right part
        pivot->next = quickSortRecur(pivot->next, newEnd);
        return newHead;
    }
public:
    Node* quickSort(Node* head) {
        Node* end = getTail(head);
        return quickSortRecur(head, end);
    }
};
```

The screenshot displays a coding platform interface with a dark theme. On the left, the 'Output Window' shows 'Problem Solved Successfully' with a green checkmark. Below this, statistics are listed: 'Test Cases Passed: 1115 / 1115', 'Attempts: Correct / Total: 1 / 1', 'Accuracy: 100%', 'Points Scored: 4 / 4', and 'Your Total Score: 134'. At the bottom, there are buttons for 'Solve Next' and 'Stay Ahead With: Build 21 Projects in 21 Days'. On the right, the code editor shows the C++ implementation of the quicksort algorithm for linked lists, with line numbers 29 to 65 visible. The code includes a private partition function and a public quickSort function.